

As the next step, consider a more general case where the covariance matrices are not equal as we have in QDA. We apply Singular Value Decomposition (SVD) to the covariance matrix of the k -th class:

$$\Sigma_k = U_k \Lambda_k U_k^\top,$$

where the left and right matrices of singular vectors are equal because the covariance matrix is symmetric. Therefore:

$$\Sigma_k^{-1} = U_k \Lambda_k^{-1} U_k^\top,$$

where $U_k^{-1} = U_k^\top$ because it is an orthogonal matrix. Therefore, we can simplify the following term:

$$\begin{aligned} & (\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \\ &= (\mathbf{x} - \boldsymbol{\mu}_k)^\top U_k \Lambda_k^{-1} U_k^\top (\mathbf{x} - \boldsymbol{\mu}_k) \\ &= (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k)^\top \Lambda_k^{-1} (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k). \end{aligned}$$

As Λ_k^{-1} is a diagonal matrix with non-negative elements (because it is covariance), we can decompose it as:

$$\Lambda_k^{-1} = \Lambda_k^{-1/2} \Lambda_k^{-1/2}.$$

Therefore:

$$\begin{aligned} & (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k)^\top \Lambda_k^{-1} (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k) \\ &= (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k)^\top \Lambda_k^{-1/2} \Lambda_k^{-1/2} (U_k^\top \mathbf{x} - U_k^\top \boldsymbol{\mu}_k) \\ &\stackrel{(a)}{=} (\Lambda_k^{-1/2} U_k^\top \mathbf{x} - \Lambda_k^{-1/2} U_k^\top \boldsymbol{\mu}_k)^\top \\ &\quad (\Lambda_k^{-1/2} U_k^\top \mathbf{x} - \Lambda_k^{-1/2} U_k^\top \boldsymbol{\mu}_k), \end{aligned}$$

where (a) is because $\Lambda_k^{-\top/2} = \Lambda_k^{-1/2}$ because it is diagonal. We define the following transformation:

$$\phi_k : \mathbf{x} \mapsto \Lambda_k^{-1/2} U_k^\top \mathbf{x}, \quad (42)$$

which also results in the transformation of the mean: $\phi_k : \boldsymbol{\mu} \mapsto \Lambda_k^{-1/2} U_k^\top \boldsymbol{\mu}$. Therefore, the Eq. (28) can be restated as:

$$\begin{aligned} \delta_k(\mathbf{x}) &= -\frac{1}{2} \ln(|\Sigma_k|) \\ &\quad - \frac{1}{2} (\phi_k(\mathbf{x}) - \phi_k(\boldsymbol{\mu}_k))^\top (\phi_k(\mathbf{x}) - \phi_k(\boldsymbol{\mu}_k)) + \ln(\pi_k). \end{aligned} \quad (43)$$

Ignoring the terms $-(1/2) \ln(|\Sigma_k|)$ and $\ln(\pi_k)$, we can see that the transformation has changed the covariance matrix of the class to identity matrix. Therefore, the QDA (and also LDA) can be seen as simple comparison of distances from the means of classes after applying a transformation to the data of every class. In other words, we are learning the metric using the SVD of covariance matrix of every class. Thus, LDA and QDA can be seen as *metric learning* (Yang

& Jin, 2006; Kulis, 2013) in a perspective. Note that in metric learning, a valid distance metric is defined as (Yang & Jin, 2006):

$$d_A^2(\mathbf{x}, \boldsymbol{\mu}_k) := \|\mathbf{x} - \boldsymbol{\mu}_k\|_A^2 = (\mathbf{x} - \boldsymbol{\mu}_k)^\top \mathbf{A} (\mathbf{x} - \boldsymbol{\mu}_k), \quad (44)$$

where $\mathbf{A}$ is a positive semi-definite matrix, i.e., $\mathbf{A} \succeq 0$. In QDA, we are also using $(\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)$. The covariance matrix is positive semi-definite according to the characteristics of covariance matrix. Moreover, according to characteristics of a positive semi-definite matrix, the inverse of a positive semi-definite matrix is positive semi-definite so $\Sigma_k^{-1} \succeq 0$. Therefore, QDA is using metric learning (and as will be discussed in next section, it can be seen as a *manifold learning* method, too).

It is also noteworthy that the QDA and LDA can also be seen as *Mahalanobis distance* (McLachlan, 1999; De Maesschalck et al., 2000) which is also a metric:

$$d_M^2(\mathbf{x}, \boldsymbol{\mu}) := \|\mathbf{x} - \boldsymbol{\mu}\|_M^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}), \quad (45)$$

where Σ is the covariance matrix of the cloud of data whose mean is $\boldsymbol{\mu}$. The intuition of Mahalanobis distance is that if we have several data clouds (e.g., classes), the distance from the class with larger variance should be scaled down because that class is taking more of the space so it is more probable to happen. The scaling down shows in the inverse of covariance matrix. Comparing $(\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k)$ in QDA or LDA with Eq. (45) shows that QDA and LDA are sort of using Mahalanobis distance.

8. LDA $\stackrel{?}{\equiv}$ FDA

In the previous section, we saw that LDA and QDA can be seen as metric learning. We know that metric learning can be seen as a family of manifold learning methods. We briefly explain the reason of this assertion: As $\mathbf{A} \succeq 0$, we can say $\mathbf{A} = \mathbf{U} \mathbf{U}^\top \succeq 0$. Therefore, Eq. (44) becomes:

$$\begin{aligned} \|\mathbf{x} - \boldsymbol{\mu}_k\|_A^2 &= (\mathbf{x} - \boldsymbol{\mu}_k)^\top \mathbf{U} \mathbf{U}^\top (\mathbf{x} - \boldsymbol{\mu}_k) \\ &= (\mathbf{U}^\top \mathbf{x} - \mathbf{U}^\top \boldsymbol{\mu}_k)^\top (\mathbf{U}^\top \mathbf{x} - \mathbf{U}^\top \boldsymbol{\mu}_k), \end{aligned}$$

which means that metric learning can be seen as comparison of simple Euclidean distances after the transformation $\phi : \mathbf{x} \mapsto \mathbf{U}^\top \mathbf{x}$ which is a projection into a subspace with projection matrix $\mathbf{U}$. Thus, metric learning is a manifold learning approach. This gives a hint that the Fisher Discriminant Analysis (FDA) (Fisher, 1936; Welling, 2005), which is a manifold learning approach (Tharwat et al., 2017), might have a connection to LDA; especially, because the names FDA and LDA are often used interchangeably in the literature. Actually, other names of FDA are Fisher LDA (FLDA) and even LDA.

We know that if we project (transform) the data of a class using a projection vector $\mathbf{u} \in \mathbb{R}^p$ to a p dimensional subspace ($p \leq d$), i.e.:

$$\mathbf{x} \mapsto \mathbf{u}^\top \mathbf{x}, \quad (46)$$

for all data instances of the class, the mean and the covariance matrix of the class are transformed as:

$$\boldsymbol{\mu} \mapsto \mathbf{u}^\top \boldsymbol{\mu}, \quad (47)$$

$$\boldsymbol{\Sigma} \mapsto \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}, \quad (48)$$

because of characteristics of mean and variance.

The Fisher criterion (Xu & Lu, 2006) is the ratio of the between-class variance, σ_b^2 , and within-class variance, σ_w^2 :

$$f := \frac{\sigma_b^2}{\sigma_w^2} = \frac{(\mathbf{u}^\top \boldsymbol{\mu}_2 - \mathbf{u}^\top \boldsymbol{\mu}_1)^2}{\mathbf{u}^\top \boldsymbol{\Sigma}_2 \mathbf{u} + \mathbf{u}^\top \boldsymbol{\Sigma}_1 \mathbf{u}} = \frac{(\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2}{\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}}. \quad (49)$$

The FDA maximizes the Fisher criterion:

$$\underset{\mathbf{u}}{\text{maximize}} \quad \frac{(\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2}{\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}}, \quad (50)$$

which can be restated as:

$$\begin{aligned} &\underset{\mathbf{u}}{\text{maximize}} \quad (\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2, \\ &\text{subject to} \quad \mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} = 1, \end{aligned} \quad (51)$$

according to Rayleigh-Ritz quotient method (Croot, 2005). The Lagrangian (Boyd & Vandenberghe, 2004) is:

$$\mathcal{L} = (\mathbf{u}^\top (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2 - \lambda (\mathbf{u}^\top (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} - 1),$$

where λ is the Lagrange multiplier. Equating the derivative of $\mathcal{L}$ to zero gives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{u}} &= 2(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \mathbf{u} - 2\lambda (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u} \stackrel{\text{set}}{=} \mathbf{0} \\ \implies (\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \mathbf{u} &= \lambda (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1) \mathbf{u}, \end{aligned}$$

which is a generalized eigenvalue problem $((\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2, (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1))$ according to (Ghojogh et al., 2019b). The projection vector is the eigenvector of $(\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1)^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2$; therefore, we can say:

$$\mathbf{u} \propto (\boldsymbol{\Sigma}_2 + \boldsymbol{\Sigma}_1)^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2.$$

In LDA, the equality of covariance matrices is assumed. Thus, according to Eq. (18), we can say:

$$\mathbf{u} \propto (2\boldsymbol{\Sigma})^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2 \propto \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2. \quad (52)$$

According to Eq. (46), we have:

$$\mathbf{u}^\top \mathbf{x} \propto (\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^2)^\top \mathbf{x}. \quad (53)$$

Comparing Eq. (53) with Eq. (23) shows that LDA and FDA are equivalent up to a scaling factor $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ (note that this term is multiplied as an exponential factor before taking logarithm to obtain Eq. (23), so this term a scaling factor). Hence, we can say:

$$\text{LDA} \equiv \text{FDA}. \quad (54)$$

In other words, FDA projects into a subspace. On the other hand, according to Section 7, LDA can be seen as a metric learning with a subspace where the Euclidean distance is used after projecting onto that subspace. *The two subspaces of FDA and LDA are the same subspace.* It should be noted that in manifold (subspace) learning, the scale does not matter because all the distances scale similarly.

Note that LDA assumes *one* (and not several) Gaussian for every class and so does the FDA. That is why FDA faces problem for multi-modal data (Sugiyama, 2007).

9. Relation to Logistic Regression

According to Eqs. (16) and (32), Gaussian and Bernoulli distributions are used for likelihood (class conditional) and prior, respectively, in LDA and QDA. Thus, we are making assumptions for the likelihood and prior, although we finally work with posterior in LDA and QDA according to Eq. (15). *Logistic regression* (Kleinbaum et al., 2002) says why do we make assumptions on the likelihood and prior when we want to work on posterior finally. Let us make assumption directly for the posterior.

In logistic regression, first a linear function is applied to the data to have $\beta^\top \mathbf{x}'$ where $\mathbb{R}^{d+1} \ni \mathbf{x}' = [\mathbf{x}^\top, 1]^\top$ and $\beta \in \mathbb{R}^{d+1}$ include the intercept. Then, logistic function is used in order to have a value in range $(0, 1)$ to simulate probability. Therefore, in logistic regression, the posterior is assumed to be:

$$\begin{aligned} \mathbb{P}(\mathcal{C}(\mathbf{x}) | X = \mathbf{x}) \\ = \left(\frac{\exp(\beta^\top \mathbf{x}')}{1 + \exp(\beta^\top \mathbf{x}')} \right)^{\mathcal{C}(\mathbf{x})} \left(\frac{1}{1 + \exp(\beta^\top \mathbf{x}')} \right)^{1 - \mathcal{C}(\mathbf{x})}, \end{aligned} \quad (55)$$

where $\mathcal{C}(\mathbf{x}) \in \{-1, +1\}$ for the two classes. Logistic regression considers the coefficient β as the parameter to be optimized and uses Newton's method (Boyd & Vandenberghe, 2004) for the optimization. Therefore, in summary, logistic regression makes assumption on the posterior while LDA and QDA make assumption on likelihood and prior.

10. Relation to Bayes Optimal Classifier and Gaussian Naive Bayes

The Bayes classifier maximizes the posteriors of the classes (Murphy, 2012):

$$\hat{\mathcal{C}}(\mathbf{x}) = \arg \max_k \mathbb{P}(\mathbf{x} \in \mathcal{C}_k | X = \mathbf{x}). \quad (56)$$